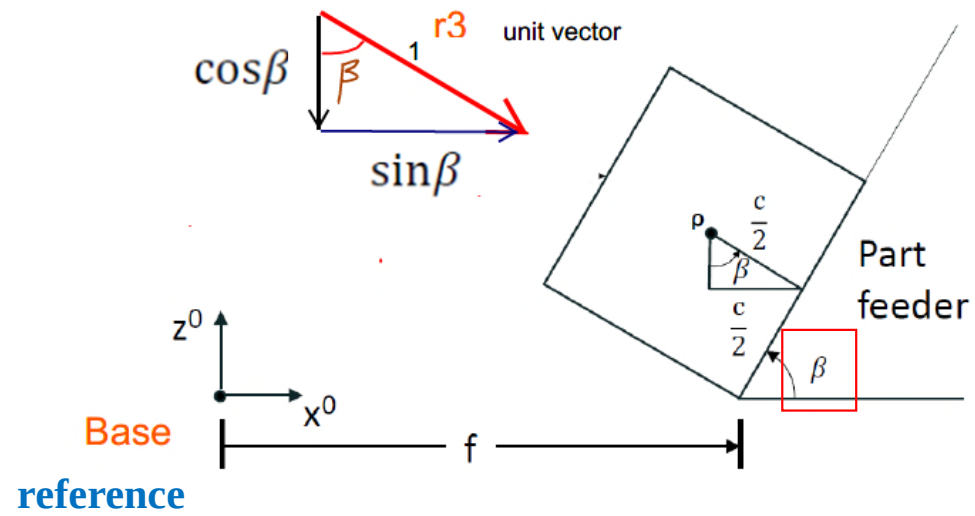


approach vector based on base coordinates obtained from DH assignments

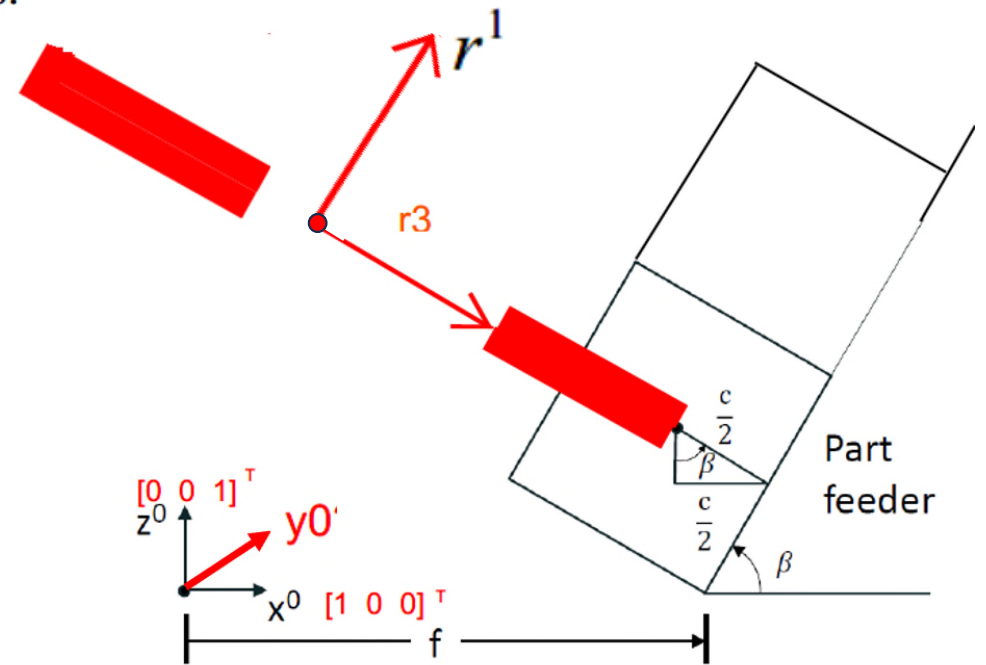


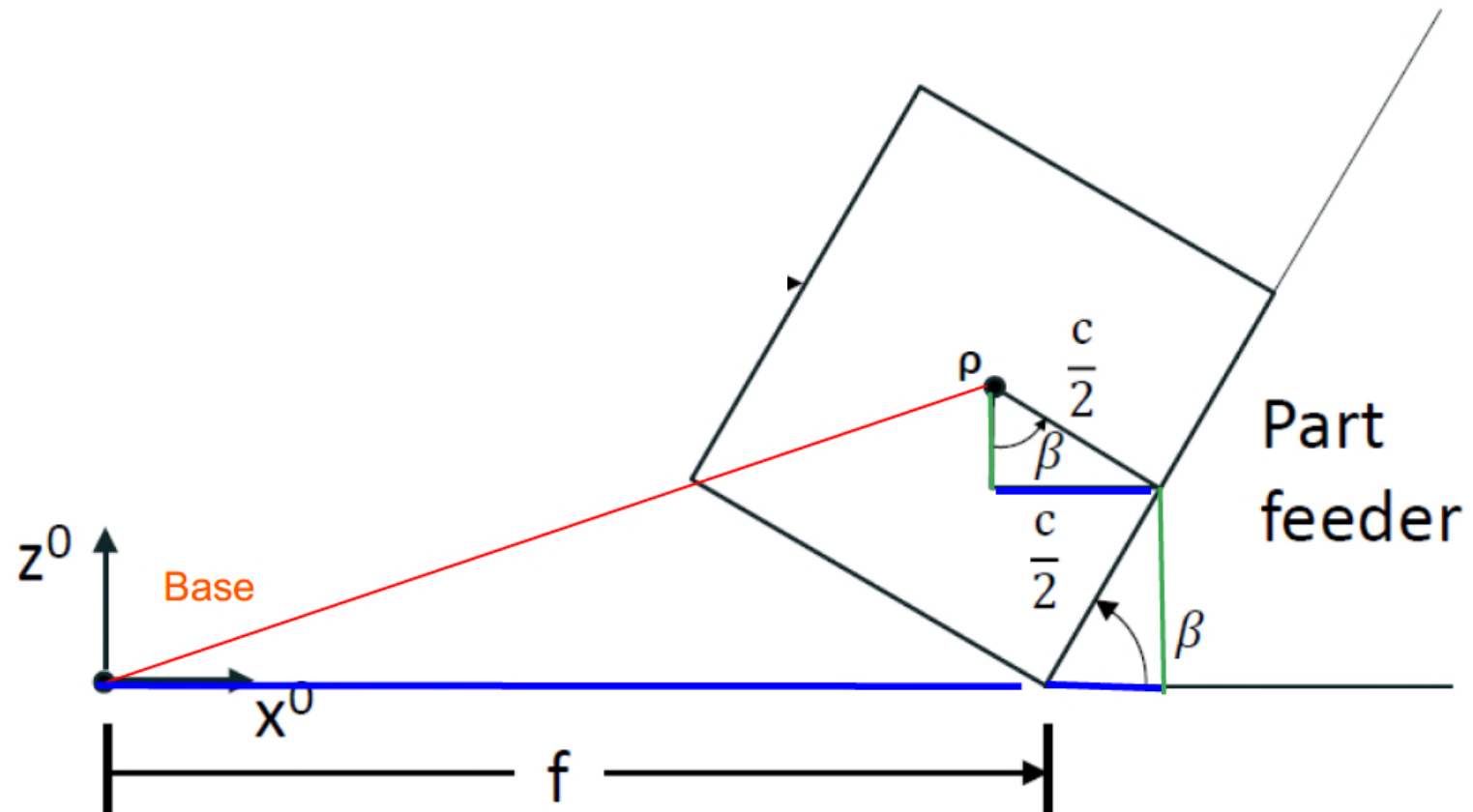
Thus the proper orientation for the tool to pick a part off the bottom of the feeder is

$$R = \begin{pmatrix} \overset{r1}{C\beta} & \overset{r2}{0} & \overset{r3}{S\beta} \\ 0 & -1 & 0 \\ S\beta & 0 & -C\beta \end{pmatrix}$$

Next consider the tool-tip position vector p . Suppose the bottom of the part feeder is located f units along the x^0 axis.

$$r^1 = r^2 \times r^3$$





$$p = [\underline{f + c(C\beta - S\beta) / 2}, 0, \underline{c(S\beta + C\beta) / 2}]^T.$$